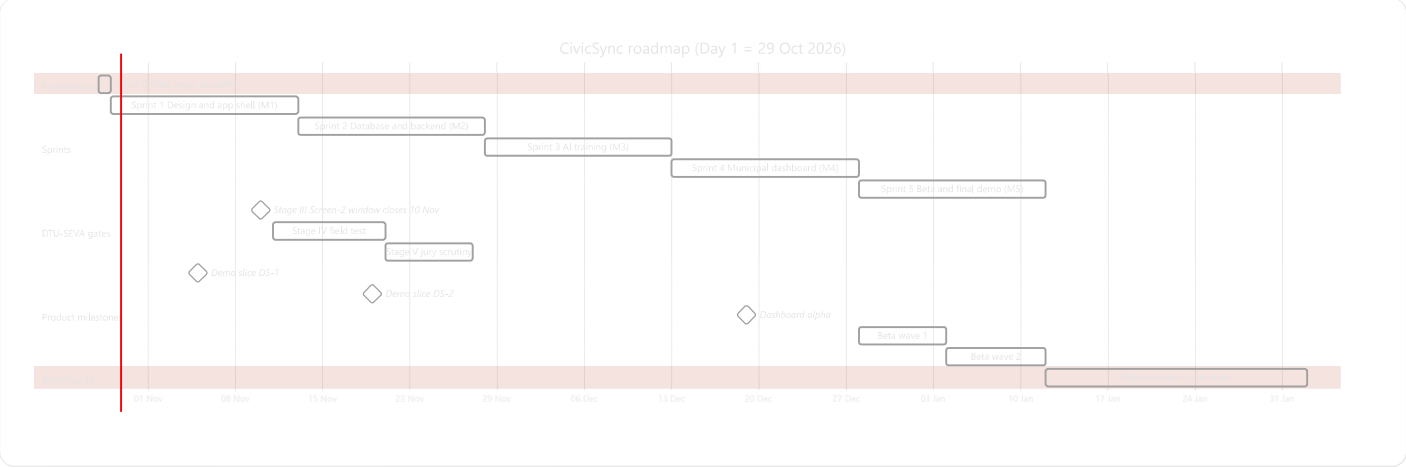
roadmap.md

> **CivicSync · Roadmap: Empty Repository to Validated Prototype and Pilot** Version 1.0-rc1 · 28 Oct 2026 · Owner: Team Leader (Aayush) One timeline. Dates assume **Day 1 = Thu 29 Oct 2026** (**A-01**). Entry and exit criteria for each phase are in `phases.md` §4; this file shows *who does what, when*.

---

1. The whole journey at a glance



---

2. Step zero: from empty repository (Day 0)

#	Task	Owner	Done when
1	Review and sign off the docs set (`loop.md`)	Team Leader	Version 1.0 tag
2	Create repository with folders: `apps/citizen_app`, `apps/dashboard`, `ml/`, `supabase/` (migrations, functions, tests), `docs/`, `.github/`	Flutter Lead	Repo exists; README explains layout
3	Branch protection, pull-request template, Dependabot, secret scanning	Backend Lead	A test push to main is blocked
4	CI that runs `flutter analyze`, `flutter test`, and Mermaid diagram checks	Flutter Lead	Green badge
5	Supabase dev project (fake data only), Firebase project, Google Cloud project with a restricted Maps key and budget alerts at 50% and 80%	Backend Lead	Keys stored as CI secrets only
6	Put the docs in `docs/`; add Vale and markdownlint	UX & Ops Lead	Docs lint runs in CI
7	Pick the three test phones (Q-10)	CV Lead	List in `docs/devices.md`
8	Confirm DTU-SEVA stage dates (Q-01) and ask Q-02	Team Leader	Answers in `memory.md`

3. The 75-day sprint plan

Sprint 1 · Design and app shell (Days 1–15, 29 Oct – 12 Nov) → M1

Days	Work	Lead
1–3	Flows and wireframes for capture, confirm, thanks, my reports; English and Hindi string files; Figma tokens from `design.md`	UX & Ops Lead
1–5	Flutter shell; camera capture; GPS with accuracy; timestamp; 18+ welcome and privacy note	Flutter Lead
1–15	Field photo collection in Agra and Jodhpur; curate public datasets; record licences; label with Label Studio or CVAT	CV Lead
5–8	Screen-2 dossier pieces; **DS-1** (Day 8); video recorded Day 7	Team Leader, UX & Ops Lead
9	**Submit Screen-2** (Fri 6 Nov); buffer to 10 Nov	Team Leader
8–14	Clickable prototype tests with 5 people; fix top 3 issues; hardware matrix tests of capture on 3 phones	UX & Ops Lead, Flutter Lead
15	**M1 go/no-go**	Team Leader

Sprint 2 · Database and backend (Days 16–30, 13 – 27 Nov) → M2

Days	Work	Lead
16–19	Schema and migrations; RLS policies and tests; private storage bucket; anonymous auth	Backend Lead
16–22	`submit_report`; ward lookup with test polygons; app submits end to end; **DS-2** (Day 22)	Backend Lead, Flutter Lead
20–26	Offline queue (drift), WorkManager sync, idempotent retries	Flutter Lead
20–28	Real ward boundaries imported (Q-04); OSM context features imported (A-12)	Backend Lead
23–29	Duplicate grouping v0 with scripted tests; Maps quotas and alerts; privacy notice draft	Backend Lead, UX & Ops Lead
23–29	(At risk) **DS-3** baseline model demo	CV Lead
30	**M2 go/no-go**: end-to-end submit, airplane-mode test, < 3 s on 4G, RLS tests green	Team Leader

Sprint 3 · AI training and integration (Days 31–45, 28 Nov – 12 Dec) → M3

Days	Work	Lead
31–35	Dataset freeze; train/test split by location; baseline training; class balance	CV Lead
36–40	Tune; INT8 quantization; **Day 40 decision**: five classes or three-class fallback	CV Lead
36–42	Severity logic and confidence rules in the app; integrate `tflite_flutter`; privacy blur v1	Flutter Lead, CV Lead
41–44	Benchmark on RD-1 and a lower-end phone; confusion matrix; model release notes	CV Lead
31–45	Priority scoring function in the database with worked-example tests (`rules.md` §1.8)	Backend Lead
45	**M3 go/no-go**: p95 < 150 ms; quality targets or fallback recorded	Team Leader

Sprint 4 · Municipal dashboard (Days 46–60, 13 – 27 Dec) → M4

Days	Work	Lead
46–52	Dashboard shell, staff sign-in with two-step check, role views, queue and ticket detail; **dashboard alpha** Day 52 (Sat 19 Dec)	Flutter Lead
46–55	`transition_ticket`, merge and split, escalation job, retention jobs; push notifications	Backend Lead
46–55	Workflow design with a ward officer walk-through (Q-03)	UX & Ops Lead
53–58	Map and density cells; overdue list; Hindi strings reviewed (Q-08); "Delete my data"	Flutter Lead, Backend Lead
55–58	Load test at 100× average; role tests; Supabase Pro on	Backend Lead
60	**M4 go/no-go**: full pipeline with a ticket through all four statuses and the citizen seeing each	Team Leader

Sprint 5 · Beta and final demo (Days 61–75, 28 Dec – 11 Jan) → M5

Days	Work	Lead
61–67	**Beta wave 1**: team plus 10 students; daily bug triage	UX & Ops Lead
68–75	**Beta wave 2**: commuters in Agra and Jodhpur; target ⩾ 40 testers, ⩾ 200 reports	UX & Ops Lead
61–75	Measure PRD-M1 to PRD-M9; timed sessions (5 per city); post-beta survey	UX & Ops Lead
66–72	Tune weights and SLA numbers from feedback; record in revision log	Backend Lead
70–75	Final demo video, results summary, docs v1.1	Team Leader
75	**M5**: validated integrated prototype, TRL 5–6, with beta results	Team Leader

4. DTU-SEVA gates and product milestones on one line

Day	Date	Event
0	Wed 28 Oct	Docs set v1.0-rc1; repo and accounts
8	Thu 5 Nov	**DS-1** ready; video recorded Day 7
9	Fri 6 Nov	**Submit Screen-2** (Stage III)
13	Tue 10 Nov	Screen-2 window closes (per screenshot)
14–22	11–19 Nov	Stage IV: field test; DS-2 due Day 22
15	Thu 12 Nov	**M1** and go/no-go
23–29	20–26 Nov	Stage V: jury scrutiny; DS-3 due Day 29 (at risk)
30	Fri 27 Nov	**M2**
40	Mon 7 Dec	Five-class or three-class decision
45	Sat 12 Dec	**M3**
52	Sat 19 Dec	**Dashboard alpha**
60	Sun 27 Dec	**M4**; dashboard beta (feature-complete)
61	Mon 28 Dec	**Closed beta** starts (wave 1)
68	Mon 4 Jan	Beta wave 2
75	Mon 11 Jan	**M5**
76+	mid-Jan onward	Pilot readiness; main DPDP duties apply from about May 2027

Stage VI (the grand finale) has no date yet. We keep the Day 75 final demo and results summary ready for it.

5. After Day 75: towards a pilot (indicative)

Step	What	When
1	Beta results review; update `rules.md` numbers; docs v1.1	Days 75–80
2	Agreement with a municipal partner; privacy notice and grievance contact live (Q-05, Q-06)	Weeks 1–3 after M5
3	One-hour staff training in one ward per city	Week 3
4	30-day ward pilot; weekly reviews with officers	Weeks 4–8
5	Pilot report and go/no-go on extending	Week 9

---

6. Critical path and risks to the schedule

Risk	Effect	Mitigation
Diwali week overlaps the Screen-2 deadline (A-02)	Rush or missed submission	Freeze content Day 6; record Day 7; submit Day 9
Stages IV and V come before M3 and M4 (C-10)	Judges see a partial build	Demo slices; honest dossier; ask Q-02
Labelled data too thin for drains and waterlogging	Model misses M3	Start collection Day 1; Day 40 fallback
Four people in four institutions, exams or holidays	Slips	Weekly reviews; one owner per task; small PRs
Free-tier pause or storage limit during demos	Outage on demo day	Keep-alive until Pro; Pro in Sprint 4
No municipal partner by Sprint 4 (Q-03)	Dashboard unvalidated	Use a mentor or retired official as a stand-in; say so in results
Hindi review not done (Q-08)	Wrong copy shipped	Don't ship unreviewed Hindi strings

Critical path: capture app (M1) → backend submit (M2) → model integration (M3) → dashboard and duplicates (M4) → beta data (M5). Data collection is the long pole that runs alongside from Day 1.

---

7. Document acceptance criteria

This document passes review when it meets the checklist for `roadmap.md` in `loop.md` §3.